

MATH255: Honours Analysis 2

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Abstract

1 Metric Spaces

1.1 Continuous functions on compact sets

2 Infinite Series

Definition 1 (Partial sum). Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. $k_0 \in \mathbb{Z}$ and $(a_k)_{k \geq k_0}$ by a sequence in X . For every $n \geq k_0$, $S_n = \sum_{k=k_0}^n a_k$ is called the partial sum of order n of $(a_k)_k$.

If $\lim_{n \rightarrow \infty} S_n = S$ in $(X, \|\cdot\|)$, for some $S \in X$, then we write

$$\sum_{k=k_0}^{\infty} a_k = S \quad (1)$$

and we say that the series $\sum_{k=k_0}^{\infty} a_k$ and that S is its sum. If $(S_n)_{n \geq k_0}$ does not converge, then we say that the series $\sum_{k=k_0}^{\infty} a_k$ diverges.

2.1 Series with nonnegative term

Theorem 1 (Direct Comparison Test). Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, \|\cdot\|)$ such that $0 \leq a_k \leq b_k$ for all $k \geq k_0$. Then

1. If $\sum_{k=k_0}^{\infty} b_k$ converges, then $\sum_{k=k_0}^{\infty} a_k$ converges and $\sum_{k=k_0}^{\infty} a_k \leq \sum_{k=k_0}^{\infty} b_k$.
2. If $\sum_{k=k_0}^{\infty} a_k$ diverges, then $\sum_{k=k_0}^{\infty} b_k$ diverges.

Corollary 1 (Limit Comparison Test). Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k > 0$ and $b_k > 0$ for all $k \geq k_0$, and $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \ell \in (0, \infty)$. Then $\sum_{k=k_0}^{\infty} a_k$ converges if and only if $\sum_{k=k_0}^{\infty} b_k$ converges.

Theorem 2 (Root Test).

Proposition 1. Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, \|\cdot\|)$ such that $a_k > 0$ for all $k \geq k_0$.

1. If $\limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$, then $\sum_{k=k_0}^{\infty} a_k$ converges.
2. If $\liminf_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1$, then $\sum_{k=k_0}^{\infty} a_k$ diverges.

Proof: (1) Assume that $R = \limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$. Then $\forall \varepsilon \in (0, 1 - R)$, $\exists n_\varepsilon \in \mathbb{N}$ such that $\forall k \geq n_\varepsilon$, $\sup_{k \geq n} \frac{a_{k+1}}{a_k} < R + \varepsilon < 1$, which gives for

all $k \geq n_\varepsilon$, $a_{k+1} < (R + \varepsilon)a_k < (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$. When $\varepsilon \in (0, 1 - R)$, $0 < R + \varepsilon < 1$, so $\sum_{k=n_\varepsilon}^{\infty} (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$ converges, which by direct comparison implies that $\sum_{k=k_0}^{\infty} a_k$ converges.

(2) Assume that $R = \liminf \frac{a_{k+1}}{a_k} > 1$. Then $\exists k_1 \geq k_0$ such that $\forall k \geq k_1$, $\inf_{k \geq n} \frac{a_{k+1}}{a_k} > R - \varepsilon > 1$, i.e. $a_{k+1} > a_k$, which implies $a_k \not\rightarrow 0$ as $k \rightarrow \infty$, thus $\sum_{k=k_0}^{\infty} a_k$ diverges. $\square$

Now we focus on series in $(\mathbb{R}, |\cdot|)$ with terms of different sign.

Theorem 3 (Alternating Series Test). *Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, |\cdot|)$ such that $0 \leq a_{k+1} \leq a_k$ for all $k \geq k_0$ and $\lim_{k \rightarrow \infty} a_k = 0$. Then $\sum_{k=k_0}^{\infty} a_k$ converges and*

$$0 \leq \sum_{k=k_0}^{\infty} (-1)^{k-k_0} a_k \leq a_{k_0} \quad (2)$$

Example 1

$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$ converges since $0 \leq \frac{1}{k+1} \leq \frac{1}{k}$ for all $k \geq 1$ and $\lim_{k \rightarrow \infty} \frac{1}{k} = 0$.

Proof: By observing that

$$\sum_{k=k_0}^{\infty} (-1)^k a_k = \sum_{k=0}^{\infty} (-1)^{k+k_0} a_{k+k_0} \quad (3)$$

$$= (-1)^{k_0} \sum_{k=0}^{\infty} (-1)^k a_{k+k_0} \quad (4)$$

we can assume that $k_0 = 0$.

For every $n \in \mathbb{N}$,

$$S_{2n} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} \quad (5)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + a_{2n} \quad (6)$$

$$\geq 0 \quad (7)$$

$$S_{2n+1} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} - a_{2n+1} \quad (8)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + (a_{2n} - a_{2n+1}) \quad (9)$$

$$\leq 0 \quad (10)$$

Definition 2 (Cauchy product). *Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be series in $(\mathbb{R}, |\cdot|)$ or $(\mathbb{C}, |\cdot|)$.*

For each $n \in \mathbb{N}$, define

$$c_n = \sum_{k=0}^n a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 \quad (11)$$

Then the series $\sum_{n=0}^{\infty} c_n$ is called the Cauchy product of $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$.

Proposition 2. If $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ are series in $(\mathbb{R}, |\cdot|)$ or $(\mathbb{C}, |\cdot|)$ such that $\sum_{k=0}^{\infty} a_k$ converges and $\sum_{k=0}^{\infty} b_k$ converges (not necessarily absolutely). Then the Cauchy product $\sum_{n=0}^{\infty} c_n$ converges and $\sum_{n=0}^{\infty} c_n = (\sum_{k=0}^{\infty} a_k)(\sum_{k=0}^{\infty} b_k)$.

Example 2 (Counter example of Cauchy product of two conditionally converging series)

Consider $a_k = b_k = \frac{(-1)^k}{\sqrt{k+1}}$. Then $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ converge by the alternating series test and p -series convergence.

In this case:

$$|c_n| = \left| \sum_{k=0}^n a_k b_{n-k} \right| \quad (12)$$

$$(13)$$

Definition 3 (Power series). Let $(a_k)_k$ be a sequence in $\mathbb{R}$ or $\mathbb{C}$ and $x_0 \in \mathbb{R}$ or $\mathbb{C}$. The function $x \mapsto \sum_{k=0}^{\infty} a_k (x - x_0)^k$ is defined at points where it converges, is called a power series.

Proposition 3 (Radius of convergence). Let $\sum_{k=0}^{\infty} a_k (x - x_0)^k$ be a power series in $\mathbb{R}$ or $\mathbb{C}$. Define $R = \frac{1}{\limsup_{k \rightarrow \infty} \sqrt[k]{|a_k|}}$ with the convention that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$. Then R is called the radius of convergence of the power series, and

$$\text{if } |x - x_0| < R, \text{ then } \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ converges,} \quad (14)$$

$$\text{if } |x - x_0| > R, \text{ then } \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ diverges.} \quad (15)$$

References